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DLRV project Proposal

Flow-Based Label Propagation for UAV Video Segmentation

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1 Introduction

1.1 Topic of This DLRV project

Semantic segmentation of UAV video is expensive because every frame requires pixel-accurate labels when annotated manually. This project implements a flow-based label propagation pipeline for a single UAV video from the Ruralscapes dataset using SEA-RAFT [1] as the optical flow backbone. Keyframe segmentation masks are warped to surrounding frames along the estimated flow field and evaluated against dense ground truth using mIoU, producing a clear picture of how label quality degrades with increasing distance from the keyframe.

1.2 Relevance of This DLRV project

In a 30 fps recording, one minute of video contains 1,800 frames. If the current workflow labels one frame per second, that is already 60 manual annotations per minute. A working propagation pipeline can extend those labels to cover many more frames with acceptable quality, directly reducing annotation cost for aerial video applications such as agricultural monitoring and infrastructure inspection.

2 Related Work

2.1 Survey of Related Work

Flow-based label propagation has been studied extensively in the context of video segmentation. The most directly related prior work is SegProp [2], which propagates sparse keyframe annotations to unlabelled aerial video frames using optical flow, demonstrating that highly redundant consecutive frames make this approach particularly effective for UAV footage. The optical flow estimation in this project relies on SEA-RAFT [1], which extends the recurrent architecture of RAFT [3] with a simplified encoder and attention module, yielding strong accuracy on standard benchmarks. The classical foundations of flow-based warping and the forward-backward consistency check used here for occlusion detection are rooted in the variational framework of Brox et al. [4]. Building on propagation as a core primitive,

Zhu et al. [5] propose a warp-and-refine paradigm in which flow-propagated labels are subsequently corrected by a lightweight network, providing motivation for the error analysis central to this project.

2.2 Limitation and Deficits in the State of the Art

Most propagation pipelines assume smooth, near-planar motion. UAV footage violates this when the drone changes altitude or flies over scenes with depth variation, introducing parallax that standard flow warping cannot fully correct [2, 5]. Furthermore, propagation quality is rarely measured as a function of frame distance, leaving practitioners with little guidance on how far from a keyframe the propagated labels remain useful [1, 3].

3 Problem Statement

This project builds a flow-based label propagation pipeline for a single densely annotated UAV video and characterises how propagation quality degrades with increasing distance from the keyframe. SEA-RAFT [1] estimates optical flow from each keyframe to surrounding frames, the keyframe mask is warped along the flow field using nearest-neighbour interpolation, and occluded pixels are detected and excluded via a forward-backward consistency check [4]. The warped masks are evaluated against ground truth using mIoU, and scores are plotted as a function of propagation distance to directly characterise pipeline reliability.

4 Project Plan

4.1 Work Packages

WP1 Literature Study and Setup. Review the core literature on optical flow and flow-based label propagation, identify key design decisions from SegProp [2], install all dependencies, and confirm that SEA-RAFT produces valid flow on at least one Ruralscapes frame pair.

WP2 Propagation Pipeline. Implement the end-to-end pipeline: run SEA-RAFT between keyframes and target frames, warp keyframe masks using nearest-neighbour interpolation, exclude occluded pixels via the forward-backward consistency check, and compute mIoU over valid pixels against ground truth. Validate on a short clip by confirming mIoU decreases with increasing frame distance.

WP3 Evaluation, Analysis, and Report. Scale the pipeline to the full video, record per-pair mIoU within a fixed propagation window, and produce a mIoU-vs-distance decay curve and a difficulty heatmap over the video timeline. Select a small set of failure cases and visualise them alongside ground-truth masks. Write the report continuously throughout the project.

4.2 Milestones

- M1 Literature reviewed; SEA-RAFT producing valid flow on one Ruralscapes frame pair.
- M2 End-to-end pipeline validated on a short clip with correct mIoU evaluation.
- M3 Full-video evaluation complete; heatmap, distance plot, failure cases, and final report submitted.

4.3 Deliverables

Minimum Viable

- A working propagation pipeline with mIoU evaluation on one Ruralscapes video.
- A mIoU-vs-distance plot and at least two annotated failure-case visualisations.

Expected

- A difficulty heatmap over the full video with a written interpretation of failure patterns.

- A per-class IoU breakdown identifying which categories propagate most and least reliably.

Desired

- A clean, documented implementation reusable on other UAV videos.
- Practical guidelines for reliable propagation distance under different motion conditions.

5 Potential Extension

If the core pipeline is completed ahead of schedule, a frame selection model may be pursued as an optional extension. SegProp [2] selects keyframes at fixed intervals, ignoring content. The per-pair mIoU scores from WP3 expose which intervals are hardest to propagate, providing a natural signal for smarter selection. The extension would build a lightweight model that automatically recommends which k frames are most valuable to annotate manually, prioritising frames at the boundary of high-difficulty intervals. The goal is to show that content-aware keyframe selection yields higher overall segmentation quality than the fixed-interval baseline of SegProp [2] under the same annotation budget.

References

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